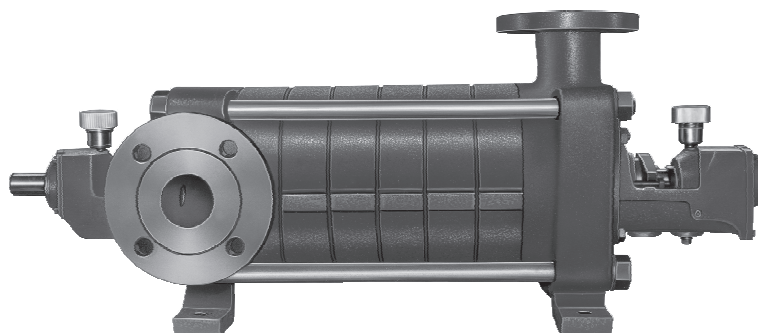


High Pressure Multistage Pumps



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0 General

Centrifugal pumps will give trouble-free, satisfactory service if they are properly installed and maintained.

Follow the instructions in this manual carefully. Do not run the pumps under operating conditions differing from those specified by us.

This manual does not take into account any site safety regulations which may apply. The site manager or site operator is responsible for notifying our erection staff of any such regulations and ensuring they are complied with.

The type series, pump size, main operating data, product no. and serial no. are all stamped on the nameplate attached to the pump. Please quote this information whenever you have queries or repeat orders and in particular when ordering spare parts. In the case of damage please contact the nearest KSB service center.



Do not operate this pump set above the limit values for capacity, speed, pressure and temperature etc. shown on the nameplate. Ensure that operation is in accordance with the instructions contained in this manual (contract documentation). It is essential that the electrical connection values are as specified and that the instructions on installation and maintenance are complied with. Operation of the pump set beyond above conditions may lead to overloads the pump set cannot withstand.

Disregard of this warning may result in personal injury and property damage.

N.B.

The instructions and descriptions in this manual refer to the standard pump model. This manual does not cover all design details or eventualities which might occur during installation, operation or maintenance. The pump set must only be handled by skilled, trained personnel.

For any information and instructions not contained in this manual, please contact KSB.

The manufacturer accepts no liability for the pump set if the instructions in this manual are not complied with.

0.1 Transport

When transporting the complete pump set attach ropes to the pump and motor as illustrated (not through the motor eyebolt).

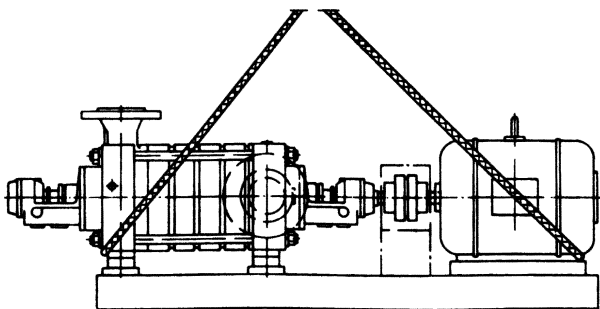


Fig. 1 Pump and motor on a common baseplate

1 Assembly / Installation at Site

1.1 Foundation

Make sure that the concrete foundation has set hard before mounting the pump set. The surface of the foundation must be completely leveled and perfectly smooth.

1.2 Installation

Position the pump set on the foundation and align using a precision spirit level (on the shaft / discharge nozzle). Ensure that the gap between the two coupling halves is as given on the installation plan. Shims should always be fitted to the left and right of the bolts in close proximity to the bolts, between the baseplate/baseframe and foundation. If the bolts are more than 800 mm apart, position extra shims equi-distant between them. All shims must lie perfectly flush.

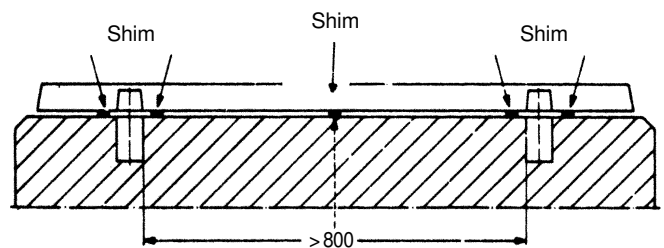


Fig. 2 Fitting shims

Uniformly tighten up the bolts.

Grout in the baseplate (non-shrinking mortar is highly recommended), ensuring no cavities remain.

1.3 Aligning the Pump and Motor



After the baseplate has been fixed in position carefully check the coupling and if necessary, realign the pump set (at the motor).

Before doing so, loosen pump feet and tighten again without transmitting any stresses and strains.

The coupling must also be checked and the pump set realigned if the pump and motor are supplied ready mounted on a common baseplate.

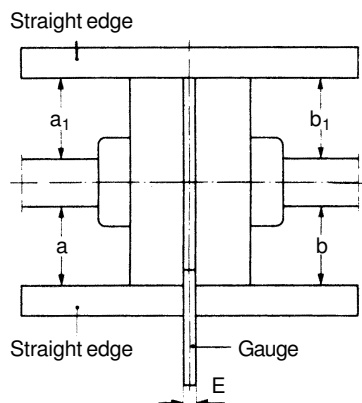


Fig. 3 Aligning a non-spacer type flexible coupling

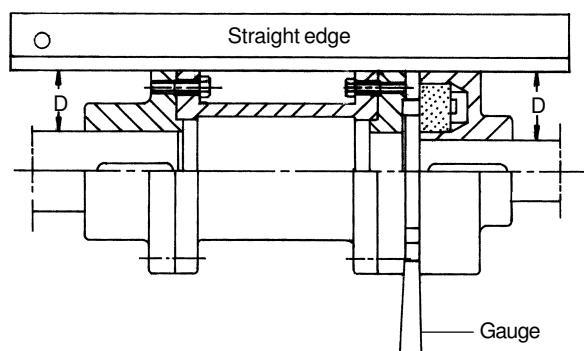


Fig. 4 Aligning a flexible spacer coupling

The axial and radial deviation between the two coupling halves must not exceed 0.1 mm.

This must also be ensured at operating temperature and under actual inlet pressure.

If the temperature of the medium handled is $\geq 100^\circ\text{C}$, the pump must be pinned to the baseplate (see Fig. 5). Tighten foundation bolts at the discharge side pump feet slightly only to facilitate free thermal expansion.

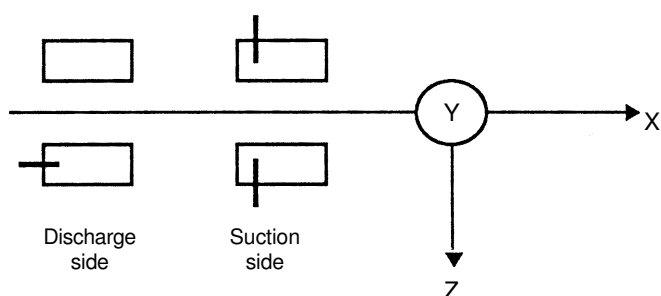


Fig. 5 Pinning the pump to the baseplate

1.4 Connecting the Piping

Never use the pump as an anchorage point for the piping.

Forces and moments of the piping may act both on suction and discharge nozzles at the same time.

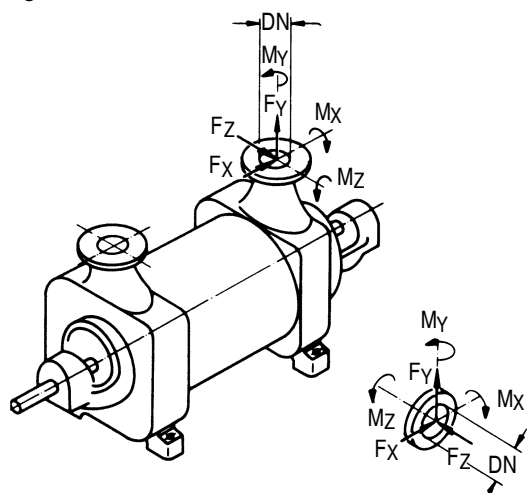


Fig. 6 Max. permissible forces and moments acting on the pump nozzles

Directions in which the forces are acting :

X = horizontal, parallel to the pump axis

Y = vertical to the pump axis

Z = horizontal at right angles to the pump axis

Directions in which the moments are acting :

MX = around the horizontal axis, parallel to the pump axis

MY = around the vertical nozzle axis

MZ = around the horizontal axis at right angles to the pump axis

DN = nominal diameter of the suction and discharge nozzle

Dimensions :

F in N

M in Nm

DN in mm

Suction and discharge nozzles have to be considered separately.

max. permissible piping forces

WKFI	40
Suction nozzle	
F _x	600
F _y [N]	1400
F _z	600
M _x	400
M _y [Nm]	300
M _z	400
Discharge nozzle	
F _x	600
F _y [N]	1400
F _z	600
M _x	400
M _y [Nm]	300
M _z	500

The suction line should be run with a rising slope towards the pump, under inflow conditions with a downward slope towards the pump.

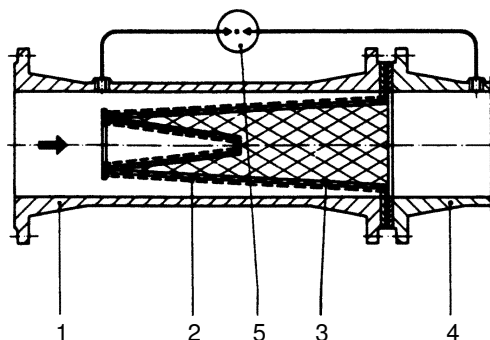
The pipes should be supported very near to the pump and should be connected to the pump without transmitting any stresses and strains to it. The pump must not bear the weight of the piping.

The nominal diameters of short pipelines shall correspond at least to those of the pump connections for long pipelines the most economic nominal diameters shall be determined depending on the respective requirements.

We recommend installing non-return devices and shut-off valves, depending on the type of plant and pump.

The thermal expansion of the pipework must be compensated for by suitable means so as not to impose any extra load on the pump.

Before commissioning a new plant, thoroughly clean, flush and blow through all vessels, piping and connections. As welding beads, scale and other impurities frequently only become dislodged after a certain period of time, it is necessary to fit a strainer in the suction line to prevent these from entering the pump. The total cross-section of the holes in the strainer should be three times the cross-section of the piping in order to avoid excessive pressure loss across the strainer due to clogging. The conical strainer consists of a coarse strainer fronted by a fine strainer with a 2.0 mm mesh and 0.5 mm diameter wire, made of corrosion-resistant material.

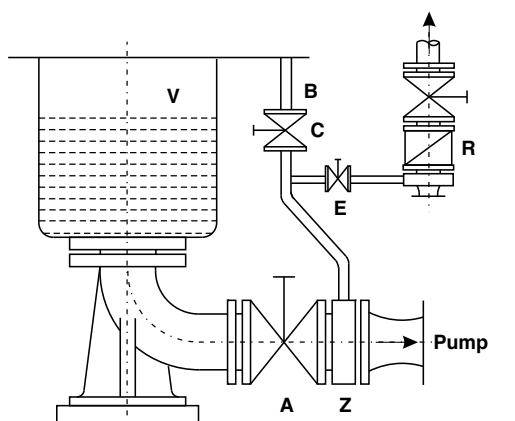


- | | |
|-------------------|-------------------------------|
| 1 Strainer holder | 4 Pump suction nozzle |
| 2 Fine strainer | 5 Differential pressure gauge |
| 3 Coarse strainer | |

Fig. 7 Conical strainer for suction line

1.4.1 Vacuum Balance Line

If the pump has to pump a liquid out of a vessel under vacuum, it is advisable to install a vacuum balance line. This line should have a nominal size of 25 mm at least. It should be arranged to lead back into the vacuum vessel at a point above the highest permissible liquid level. An additional line starting at the pump discharge nozzle facilitates venting of the pump before start up. The vacuum-tight isolating valve E in this connecting line should be closed after the venting procedure and should remain closed while the pump is running. The main isolating valve C in the vacuum balance line must remain open at all times when the pump is running and should be closed when the pump is shutdown. Refer fig. 8.



- | | |
|--------------------------------|-----------------------|
| A Main isolating valve | R Check valve |
| B Vacuum balance line | V Vacuum vessel |
| C Isolating valve | Z Intermediate flange |
| E Vacuum-tight isolating valve | |

Fig. 8 Suction line and vacuum balance line

1.4.2 Minimum Flow

If it is possible that the pump will have to operate against a closed shut-off valve arranged in the discharge line, a minimum flow of

- | | | |
|-----|-----------------|----------------------|
| t | -10 to +100°C | = 15% of Q_{opt} . |
| t | > 100 to +140°C | = 20% of Q_{opt} . |
| t | > 140 to 200°C | = 25% of Q_{opt} . |

must be provided during that time.

If special cases require a detailed calculation, contact KSB.

1.5 Checking the Direction of Rotation

The rotational direction must correspond to the arrow on the pump. Check this by switching the pump on and then off immediately.

Always check the direction of rotation with the motor disconnected.

1.6 Coupling Guard

Safety regulations specify that the pump must be fitted with a coupling guard. If the purchaser stipulates that he does not want us to supply a guard, then this must be provided by the operator.

1.7 Final Check

Re-check alignment as described in 1.3. It must be possible to rotate the coupling easily by hand.

Check if all connections are correct and function properly.

2. Start-up / Commissioning, Shutdown

2.1 Preparations for Start-up

2.1.1 Lubricants

The bearing bracket should be filled with oil of any one of the following types and specifications :

Indian Oil	:	Servosystem 46
Hindustan Petroleum	:	Enklo 46
Bharat Petroleum	:	Bharat Hydrol 46

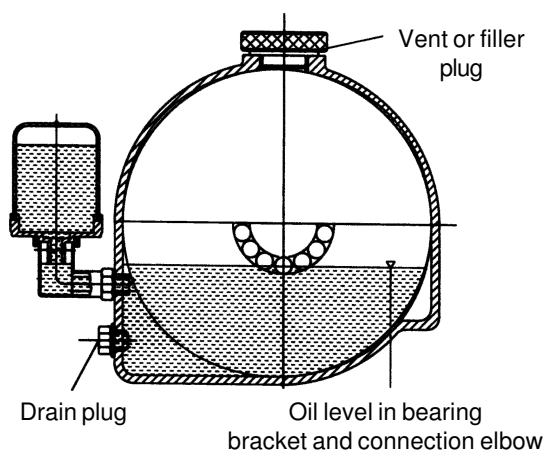


Fig. 9 Oil fill

Procedure :

Unscrew vent plug. Pour in oil through the vent plug aperture after removing the reservoir of the constant level oiler, until oil appears in the vertical portion of the connection elbow of the constant level oiler (see fig. 9).

Then fill the reservoir of the constant level oiler with oil and fit it back into operating position. Screw vent plug in again. After a short time has elapsed, check whether the oil level in the reservoir has sunk. The reservoir should always remain filled. If the vent plug is inaccessible or difficult to reach e.g. the oil can be filled into the bearing bracket through the connection elbow of the constant level oiler.

Caution :

The oil level should always be situated below the level of the vent slot arranged at the top edge of the connection elbow and this slot should always be perfectly dry. Do not tighten the elbow by applying force on the reservoir use lock nut for this purpose.

2.1.2 Shaft Seal

Shaft seal (see 4.3.2)

2.1.3 Priming the pump and Associated Checks

Pumps and suction line must be vented and filled with the medium handled prior to start-up. The shut-off valve in the suction line must be fully open.

Fully open all auxiliary connections and check the flow. Open shut-off valve in the vacuum balance line (if fitted) and close vacuum-tight shut-off valve E (Fig. 9).

Open minimum flow line of the non-return valve (if fitted). Secure minimum flow line against accidental closing.

Non-return valve with hand-operated line, open valve of this line. If the pump is fitted with a hand-operated minimum flow line (by-pass) then open this line. If a check valve is fitted and start-up is to be effected with shut-off valve open the check valve must be closed under the counter-pressure (e.g. boiler pressure). If prior to start-up there is not sufficient counter pressure then start-up with closed shut-off valve.

2.2 Start-up

Start the set up only if the shut-off valve in the discharge line is closed. Once the pump has reached full speed slowly open the valve and set to the duty point.

N.B.

After the pump has reached its operating temperature and/or if the pump leaks tighten the tie bolts with the pump set shut-down.

2.3 Shutdown

Close the shut-off valve in the discharge line.

If a check valve is installed in the discharge line, the shut-off valve can remain open as long as there is a counter-pressure.

Switch off the motor making sure the pump set runs smoothly and evenly down to a standstill.

If the pump set is to remain out of service for long periods, close the shut-off valve in the suction line. Close off the auxiliary connections.

On pumps supplied with medium under vacuum, sealing liquid must be supplied to the stuffing box even during standstill.

If there is a danger of freezing and / or if the pump is to be out of service for long periods, drain the pump or otherwise protect against freezing.

3. Maintenance and Lubrication

3.1 Supervision of Operation

The pump must run quietly and evenly at all times.

Max. permissible room temperature 40°C.

The bearing temperature may exceed the room temperature rise by up to 50°C, but must not go above 90°C (measured externally on the bearing housing).



Never allow the pump to run dry.

Do not run the pump against a closed shut-off valve for a prolonged period.

The shut-off valves in the auxiliary feed lines must not be closed during operation.

If the pump has gland packing this should leak slightly during operation. The gland cover should only be lightly tightened.

In case of prolonged periods of standstill, the pump should be started up and then shut down immediately to ensure it is in constant readiness for operation.

- Fire fighting water	1 start-up/shutdown/month
- Drinking water	1 start-up/shutdown/48 hrs
- Standby pumps	1 start-up/shutdown/week

Check proper functioning of the auxiliary connections.

The flexible coupling elements should be regularly checked and replaced as soon as they show signs of wear.

3.2 Lubrication and Grease / Oil Change

3.2.1 Lubrication

The antifriction bearings are lubricated by means of oil. For required quantities see 4.2.2.3.

3.2.2 Oil Change

First oil change after 300 operating hours, all further changes every 3000 operating hours.

Procedure :

Remove threaded plug below bearing housing 350.1 and drain oil. After draining of the bearing housing screw in the plug again and fill in oil as described under 2.1.1.

4. Special Instructions and Recommendations

4.1 Basic Instructions / Recommendations



Before dismantling, make sure the pump set is disconnected from the power supply and cannot be switched on accidentally.

The suction and discharge shut-off valves must be closed.

Make sure that the pump casing has cooled down to ambient temperature and that it is empty and not under pressure.

4.2 Dismantling

1. Drain oil as described under 3.2.3 if oil lubrication is provided.
2. Dismantle auxiliary connections.
3. Remove coupling guard.
4. If coupling is not fitted with intermediate sleeve : Disconnect pump from motor and detach from baseplate.
5. If coupling is fitted with intermediate sleeve : For work on the shaft seals, bearings or shaft protecting sleeves or shaft sleeves respectively, the pump may remain on the baseplate and connected to the pipeline.

If the pump has been in operation for a long time it may be difficult to remove certain parts from the shaft. If this is the case, use a brand name penetrating oil or suitable pull-off device.

Never use force !

4.2.1 Gland Packing

Remove hex. nut 920.2, gland cover 452, gland packing 461, lantern ring 458.

4.2.2 Bearings

Angular Contact Ball Bearing

Remove hex. head bolt 901.1, end bearing cover 361.2, spacer disc 551.1, withdrawal nut (920), lock washer (931). Press bearing casing 350.2 off discharge casing 107 and withdraw with centering sleeve 525 angular contact ball bearing 320, spacer disc 551.1, spacer ring 504.1, gasket 411.4.

4.2.2.1 Movable Bearing (motor side)

Withdraw coupling half using a pull-off device : Remove (gasket 411.4), hex. head bolt 901.1, bearing cover 360, spacer disc 551.2 loosen clamping sleeve 531, hex. nut 920.1. Withdraw bearing housing 350.1 with cylindrical roller bearing 322 and clamping sleeve 531 (WKFI 40 - 100) from suction casing 106. Remove gasket 411.4.

4.2.2.2 Bearing Sizes

WKFI	Stationary Bearing Angular contact ball bearing DIN 628	Movable Bearing Cylindrical rotor bearing with clamping sleeve DIN 5412 DIN 5415	
40	7305 BUA	NU 206 K C3	H 206

4.2.2.3 Lubrication Quantities

WKFI		Code	Oil lubrication Oil per bearing in Ltrs.
40	Stationary Bearing	7305 BUA	0.18
	Movable Bearing	NU 206 K C3	0.16

4.2.3 Pump Body

Always dismantle the pump from the discharge side.

Stacken hex. nuts 920.3 allowing a slight pretension to remain.
Loosen pump feet attachment screws.

Lift pump off the baseplate.



Do not handle with ropes attached to the pump shaft.

Fully loosen hex. nuts 920.3. Remove disc 550.1 withdraw tiebolts 905.



Place wedge or frame under stage casing 108.1 so as to allow free access to the parts to be dismantled.

Prior to dismantling position the stage casings 108.1 against each other and number consecutively to ensure when assembling suction casing 106, stage casing 108.1 and discharge casing 107 are fitted again in the same order and position.

Remove discharge casing 107, casing wear ring 502.2, O-ring 412.2 and diffuser 171.3 at the same time : if not yet dismantled, withdraw circlip 932.1 and shaft protecting sleeve 524.1 (shaft sleeve 523.1, hex nut 920.4, spacer sleeve 525.2). Dismantle in the following order : stage casing 108.1, O-ring 412.1, casing wearing 502.1/2, diffuser 171.1/2, impeller 230.1/2.

Store stage casing in such a way that damage to the sealing surfaces is avoided.

4.2.4 Shaft

Check for true running on turning lathe.
Max. admissible out-of-truth tolerance 0.03 mm.
Re-conditioning of shaft 210 not allowed

4.2.5 Rotor Clearance

		New		max. admissible	
		Cast iron bronze	Chrome steel	Cast iron bronze	Chrome steel
		mm in Ø		mm in Ø	
Casing wear ring impeller neck		0.30	0.35	0.80	0.80
Diffuser impeller hub			0.40		
Throttle sleeve / throttling bush			0.35	2.00	2.00
Shaft / suction casing	Suction lift operation				
	Suction operation			0.80	0.80

4.2.6 Pump Rotor



Mount impeller 230.1/2 (keywayed nut 920.4, spacer sleeve 525.2 or throttle sleeve 522), shaft protecting sleeve 524.1 / shaft sleeve 523.1, adjusting discs 550.2, circlip 932.1 and pump side coupling half onto shaft 210.

4.3 Reassembly

4.3.1 Pump

Reassemble in accordance with standard engineering practice. Coat fits of individual components and screw connections with graphite or similar prior to assembly.

Check the O-rings for wear and replace if necessary.

Reassemble the pump in reverse order to dismantling and ensure the components are fitted in the correct sequence Preheat bush 540.2 prior to assembly. When mounting new parts align pump rotor in the casing center with respect to the existing axial clearance. For this purpose displace the rotor - the pump being assembled but the (discharge side) stationary bearing dismantled - first to the suction side then to the discharge side to its stop and take the axial measurements. From both measurements obtain the center position and fix with adequate number of spacer discs 551.1 (see general drawing page 12).

4.3.2 Shaft Seal

4.3.2.1 Stuffing Box Compartment

Thoroughly clean the packing compartment and shaft protecting sleeve before packing the gland.

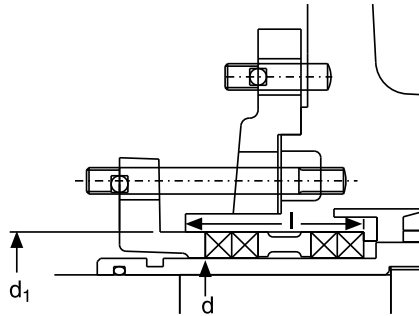


Fig. 11 Stuffed box compartment with neck ring

mm	d	d ₁	l
Packing compartment	35	51	54
No. of rings	4 packing rings 1 lantern ring		

4.3.2.2 Packing Ring



Fig. 12 Packing ring cut to size

Packing required on each side :

WKF	40
Cord thickness s mm	8 ²
Cord length l mm	≈ 580

Insert first packing ring and push it home using the gland cover.

Fit each subsequent ring separately with its joint offset by 90° to the preceding one pushing it home with the gland cover. Put L. ring as per the position shown.

Tighten the gland cover slightly and uniformly. It must be possible to rotate the rotor without difficulty.

4.3.2.3 Packing Ring of Aramid

Before cutting the packing cord wrap the cut faces with PTFE (polytetrafluorethylene) tape and cut in vertical direction to the longitudinal axis of the cord. The PTFE tape must not be removed from the cut faces again.

Insert lantern ring as well as the first packing ring with cut face, press straight inwards and tighten slightly only.

Fit each subsequent packing ring with its joint offset by approx. 90° to the preceding one, pushing it home with the gland cover.

Tighten the gland cover by hand and subsequently loosen by ½ turn. Use thickness gauge to check uniform seat of the gland cover under pressure. Set leakage during approx. 30 minutes while the pump is operating.

Leakage value : ≈ 2 - 2.5 l/h

If the leakage value is too low, the nuts at the gland cover must be loosened.

4.3.2.4 Stuffing Box Cooling

Use cleaned cooling water without the tendency to precipitate. Difference in temperature between inlet and outlet ≈ 15 °C. Max. admissible outlet temperature 80 °C. Pressure up to 10 bar.

Water cooling from external source with medium handled > 70°C.

For cooling water required see Fig. 14.

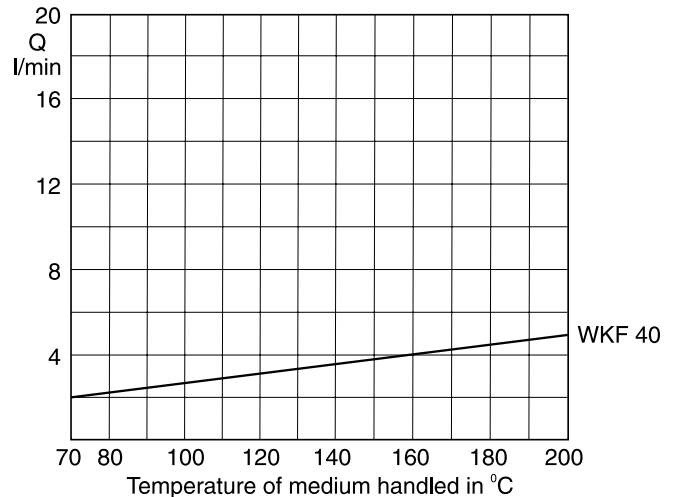


Fig. 13 Cooling water required for jacket cooling

4.4 Tightening Torques of the Tiebolts

Material	Torque Nm
EN 8	70
EN 24 V	80

4.5 Switching Frequency

Reference value max. 30 switchings/h

4.6 Speeds

Max. admissible speed n = 1/min as per fig. below.

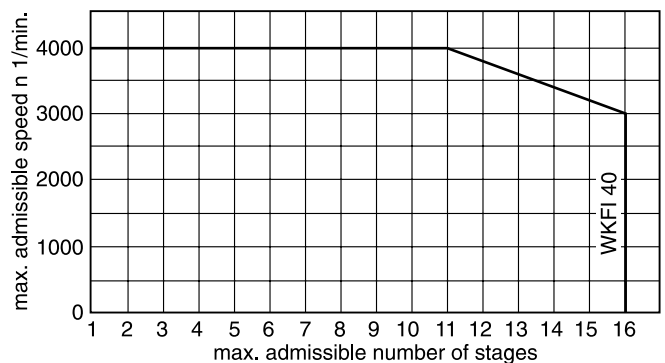


Fig. 14 Max. admissible speed

4.7 Spare Parts

4.7.1 Ordering Spare Parts

When ordering spare parts please always indicate the following information, which may be taken from the nameplate of the pump :

Type : WKFI Pump size / No. of stages (e.g. 40/5)

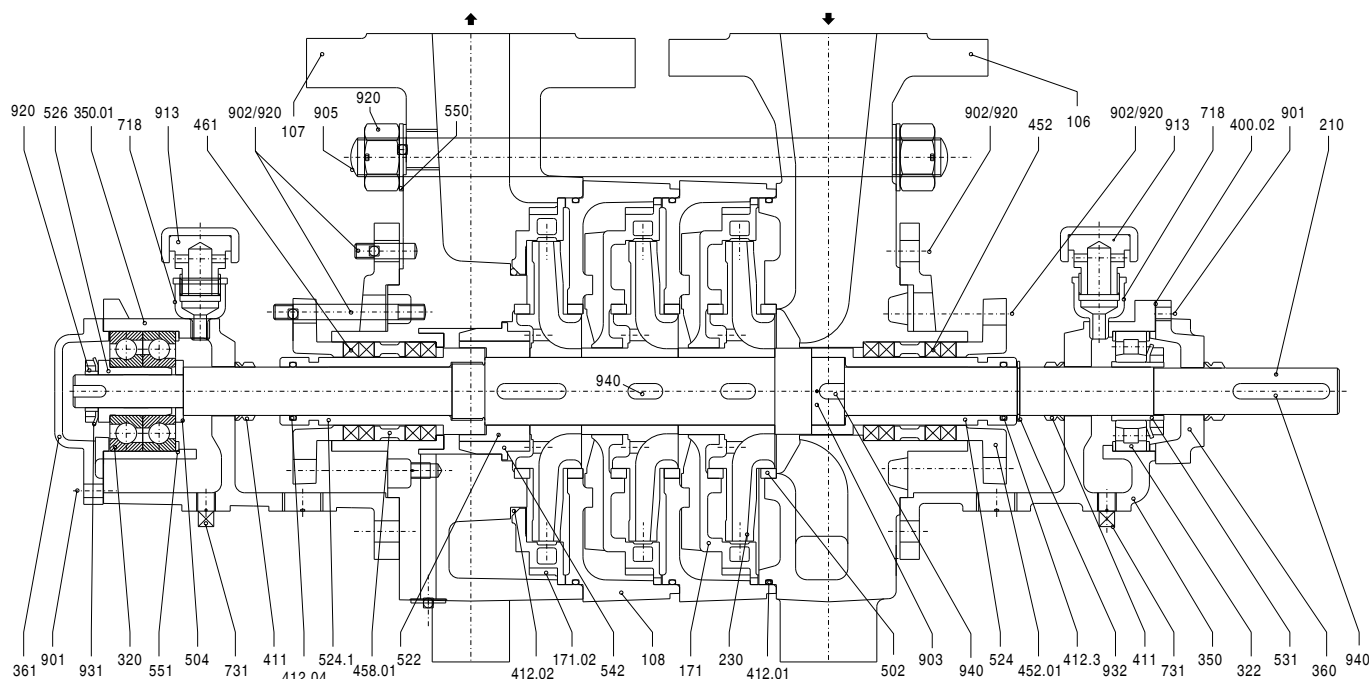
Spare part No.:

No.:

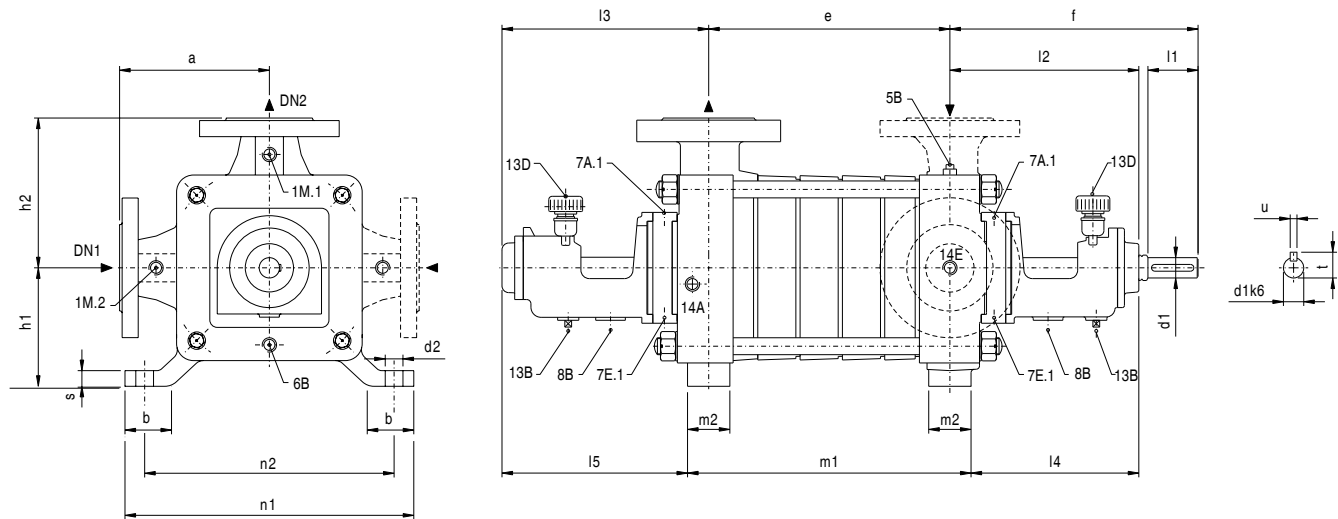
4.7.2 Recommended stock of spare parts for 2 years continuous operation

Part No.	Part Description	No. of pumps incl. standby pumps							
		2	3	4	5	6, 7	8, 9	10 & >10	
210	Shaft with Keys	1	1	2	2	2	3	30%	
230	Impellers (set)	1	1	1	2	2	3	30%	
320	Angular Contact Ball Brg. (set)	1	1	2	2	3	4	50%	
322	Cylindrical Roller Bearing	1	1	2	2	3	4	50%	
411	Gasket	2	2	4	4	4	6	50%	
412	O Ring (set)	4	6	8	8	9	12	150%	
461	Stuffing Box Packing	4	4	6	6	6	8	40%	
502.1/2	Wearing Rings (2 x S)	2	2	2	3	3	4	50%	
504	Spacer Ring	1	1	2	2	3	4	50%	
524.1	Shaft Protection Sleeve	2	2	2	3	3	4	50%	
526	Centering Sleeve	1	1	2	2	3	4	30%	
551.1	Spacer Washer	1	1	2	2	3	4	50%	

5. Sectional drawing and list of components



Part No.	Part designation	Part No.	Part designation
106	Suction casing	522	Throttle sleeve
107	Discharge casing	524.01	Shaft prot. sleeve
108.01	Stage casing	524.02	Shaft prot. sleeve
108.02	Stage casing	526	Centering ring
171.01	Diffuser	531	Adaptor sleeve
171.02	Diffuser / Last stage	542	Throttle bush
210	Shaft	550	Washer
230.01	Impeller stage	551	Spacer disc
230.02	Impeller / Last stage	718	Socket
320	Angl. cont. ball bearing	731	Square head plug
322	Cylin. roller bearing	901	Hex. bolt
350.01	Bearing housing / Suc.	902.01	Stud
350.02	Bearing housing / Dis.	902.02	Stud
360	Bearing cover	903	Hex. head plug
361	Bearing cover	905	Tie rod
400.02	Gasket	913	Vent plug
411.01	V-Ring	920.01	Hex. nut
411.02	V-Ring	920.02	Hex. nut
412.01	O-Ring	920.03	Hex. nut
412.02	O-Ring	920.04	Slotted nut
412.04	O-Ring	932	Circlip
452	St. Box Housing	940.01	Key
458	Lantern ring	940.02	Key
461	Gland packing	940.03	Key
502	Wearing ring	940.04	Key
504	Spacer ring		



WKFI	DN1	DN2	a	b	d2	f		h1	h2	l1	l2		l3		l4		l5		m2	n1	n2	d1	s	t	u
						N	HWD				N	HWD	N	HWD	N	HWD	N	HWD							
40	50	40	185	60	16	264	293	140	185	60	194	223	215	244	169	198	190	219	50	300	250	24	20	27	8

	NO. OF STAGES															
	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16
e	80	135	185	235	285	335	385	435	485	535	585	635	685	735	785	835
m1	130	185	235	285	335	385	435	485	535	585	635	685	735	785	835	885

Conncion	Designation	WKFI40
1 M.1/2	Pressure gauge	G 1/2"
5 B	Vent connection	G 1/4"
6 B	Casing Drain	G 1/4"
7 A.1/7 E.1	Cooling water outlet / inlet	G 1/4"
8 B	Leakage drain	G 1/2"
13 B/13 D	Oil drain / Topping up - venting	G 1/8"
14 A / 14 E	Balancing liquid outlet / inlet	G 1/4"

6. Faults

Pump delivers insufficient liquid	Motor overloaded	Pump does not deliver	Increased bearing temperature	Pump leaks	Excessive shaft seal leakage	Pump runs noisily and roughly	Excessive temperature increase in the pump	Cause	Remedy ¹⁾
•		•						Pump generates an excessively high differential pressure	Reset duty point Check plant for contamination Fit larger impeller(2) ²⁾ Increase speed (of turbine or combustion engine)
•		•				•		Pump and / or piping inadequately vented or primed	Vent or prime
•		•						Suction line or impeller(s) clogged	Remove deposits in the pump and / or piping
•		•						Formation of air pockets in the piping	Alter the piping layout Fit a vent valve
•		•			•	•	•	Suction head too high / NPSH pump - is too low	Correct the liquid level Fully open the suction side shut-off valve Modify the suction line if the piping losses are excessive Check the suction strainers / suction aperture Ensure that the permissible rate of pressure decrease is not exceeded
•		•						Ingress of air at the shaft seal	Clean out the sealing liquid passage, possibly introduce an external source sealing liquid or increase the pressure Fit a new shaft seal.
•		•			•			Reverse rotation	Change over two phases of the power supply
•		•						Speed too low ²⁾	Increase the speed
•	•		•			•		Excessive wear of pump internals	Replace the worn components
	•					•		Pump back pressure is lower than specified in purchase order	Reset the duty point correctly
•	•							Specific gravity or viscosity of the medium handled is higher than specified in the purchase order	²⁾
	•				•		•	Gland cover is too tight or tightened askew	Rectify, allow larger leakage quantity
	•							Speed too high	Reduce the speed ²⁾
				•				Tiebolts / gaskets	Tighten up the tiebolts Fit new gaskets
					•			Worn shaft seal	Fit a new shaft seal Check the pressure of the flushing / sealing liquid
					•			Surface of the shaft protecting sleeve / shaft sleeve is scored or rough	Fit a new shaft protecting sleeve / shaft sleeve Fit a new shaft seal
					•			Uneven running of pump	Rectify the suction conditions Realign the pump Rebalance the rotor Increase the pressure at the suction nozzle
			•		•	•		Pump set misaligned	Realign

1) The pump should be made pressureless before attempting to remedy faults in parts under pressure. 2) Please contact KSB

Pump delivers insufficient liquid	Motor overloaded	Pump does not deliver	Increased bearing temperature	Pump leaks	Excessive shaft seal leakage	Pump runs noisily and roughly	Excessive temperature increase in the pump	Cause	Remedy ¹⁾
			•		•	•		The pump is warped or there are resonance vibrations in the piping	Check the piping connections and the pump holding bolts, if necessary, reduce the gap between the pipe clamps Fix piping over vibration absorbing material
			•					Excessive axial thrust ²⁾	Clean out the balance holes in the impeller Check impeller tolerances and axial impeller adjustment
			•			•		Excessive, insufficient or unsuitable lubricant	Reduce or increase the lubricant quantity or change the lubricant
			•					Specified coupling gap not respected	Correct the coupling gap as per the installation\ plan
•	•							Motor runs on two phases	Replace the defective fuse Check the electrical connections
			•		•	•		Rotor unbalanced	Rebalance the rotor
			•		•	•		Bearings defective	Fit new bearings
			•			•	•	Insufficient flow rate	Increase the minimum flow rate
					•			Faults in the circulation liquid supply	Increase the cross-section of the line supply
•	•	•				•		Check valve defective or not fitted for operation in parallel	Check
					•			Stuffing box packed incorrectly, wrong packing material	Check
			•		•		•	Stuffing box space cooling defective	Check cross-section of cooling liquid supply
•			•		•			Change in the cross section of the balance liquid recirculating line wear in the balance device	Check balance line Check clearance at shaft in the discharge casing

1) The pump should be made pressureless before attempting to remedy faults in parts under pressure.

2) Please contact KSB

Annexure I - Environment Protection - Product Disposal after useful life

Products manufactured by KSB Pumps are designed with utmost care for environmental protection. Innovative designs and wide product range takes care of specific customer requirements reducing material and electrical energy consumption. Product materials are recyclable. Our customers are educated with environment friendly methods of disposing product ingredients at the end of their useful life. Please find herewith methods of disposing used pump ingredients.

Environment Protection measures during product disposal

Sr.	Product	Ingredients	Disposal Methods
1.	Pumps	Paint : Pump body, Base frame	Sludge to be disposed through authorised re processor
		Non ferrous parts : a) Impellers, Diffusers b) Bushes c) Stage sleeves d) Thrust plate e) Wearing ring f) Coupling Guard g) Brg. Guard	To be disposed through authorized re-processor
		Rubber / Plastic Parts : a) Cover for NRV b) Plastic cap c) Nozzle blanking d) Cooling Fan	To be disposed through authorized re-processor
2.	Motor	Paint : Motor body, Rotor	Sludge to be disposed through authorised re-processor
		Non ferrous parts : a) Support bushes b) Copper sticks and stampings c) Balancing rings d) Winding wire with varnish	To be disposed through authorized re-processor
		Rubber / Plastic Parts : a) Oil seal b) Sand guard c) Notch keys, slot insulators d) Overhang protection pipe	To be disposed through authorized re-processor

KSB Pumps and Valves - We are where you are

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Irrigation & Process Division
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 Pune - 411 018.
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Power Projects Division
 D-II Block, MIDC Chinchwad,
 Pune - 411 019.
 Tel.: 020-2740 9100 Fax : 020-2747 0890

Coimbatore :
Valves Division
 151, Mettupalayam Road,
 NSN Palayam Post,
 Coimbatore - 641 031.
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Sinnar :
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 Nashik - 422 113.
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Vambori :
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 Vambori, Ahmednagar - 413 704.
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 126, Maker Chambers III, Nariman Point,
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 Gondwara Road, Raipur - 493 221.
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 4-B, Ramkrishna Chamber,
 Productivity Road, Baroda - 390 005.
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 N5/39, 1st Floor, IRC Village, Nayapalli,
 Bhubaneswar - 751 015.
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